

What is the **first** and foremost step in machine vision system?

☐ Image enhancement

☐ Segmentation

☒ Image acquisition

☐ Classification

Intensity levels in 5-bits image are

$$2^5 = 32$$

☐ Intensity levels=64

☒ Intensity levels=32

☐ Intensity levels=256

☐ Intensity levels=128

Digitizing the image **coordinate** values is called

☐ radiance

☐ quantization

☐ luminance

☐ **sampling**

In Binary image of 30 rows and 40 columns ,
calculate the bytes required to store the image .

$$30 \times 40 = 1200$$

$$1200 / 8 = 150$$

☐ The bytes required=1000

☒ The bytes required=150

☐ The bytes required=120

☐ The bytes required=1200

What is the next step in machine vision system after Segmentation?

☐ Classification

☐ Image acquisition

☒ Feature Extraction

☐ Preprocessing

Suppose we have a 1D image with values as $A = [2, 1, 3, 0, 2]$. Now we apply the Average filter on this image of size 1x3. What would be the value of pixel A(2)?

$$2+1+3=6$$

$$6/3=2$$

1 $\rightarrow$ 2 increase by 1

☐ The value would remain the same

☐ The value would increase by 2

☐ The value would decrease by 1

☒ The value would increase by 1

Suppose we have a 1D image with values as $A = [2, 1, 5, 1, 3]$. Now we apply the Median filter on this image of size 1×3 . What would be the value of pixel $A(4)$?

1 3 5

1 $\rightarrow$ 3 increase by 2

☐ The value would remain the same

☐ The value would decrease by 1

☐ The value would increase by 1

☒ The value would increase by 2

Find the size of an image with dynamic range [0, 127] and resolution = 1600

☒ Image size= 1400 Bytes

128 $\rightarrow 2^7$

7 bit

$1600 \times 7 = 11200$ bits

$11200 / 8 = 1400$ bytes

☐ Image size= 1000 Bytes

☐ Image size= 1600 Bytes

☐ Image size= 1200 Bytes

The number of bits required to represent a pixel in an image with dynamic range $[0, 80]$ is

☐ number of bits=7

☐ number of bits=5

☐ number of bits=6

☐ number of bits=8

In machine vision system, what is the **next** step after **Image acquisition** ?

☐ Feature Extraction

☒ Preprocessing

☐ Classification

☐ Segmentation